

- 10** **C** Apply Newton 2nd law:
Along tangent of circular path,
 $F - mg \sin \theta = 0$

Along the radial direction,

$$T - mg \cos \theta = mr\omega^2$$

Thus $T = mr\omega^2 + mg \cos \theta$, so T varies with $\cos \theta$

Note: Tension is largest at lowest point and smallest at highest point in a vertical circle.